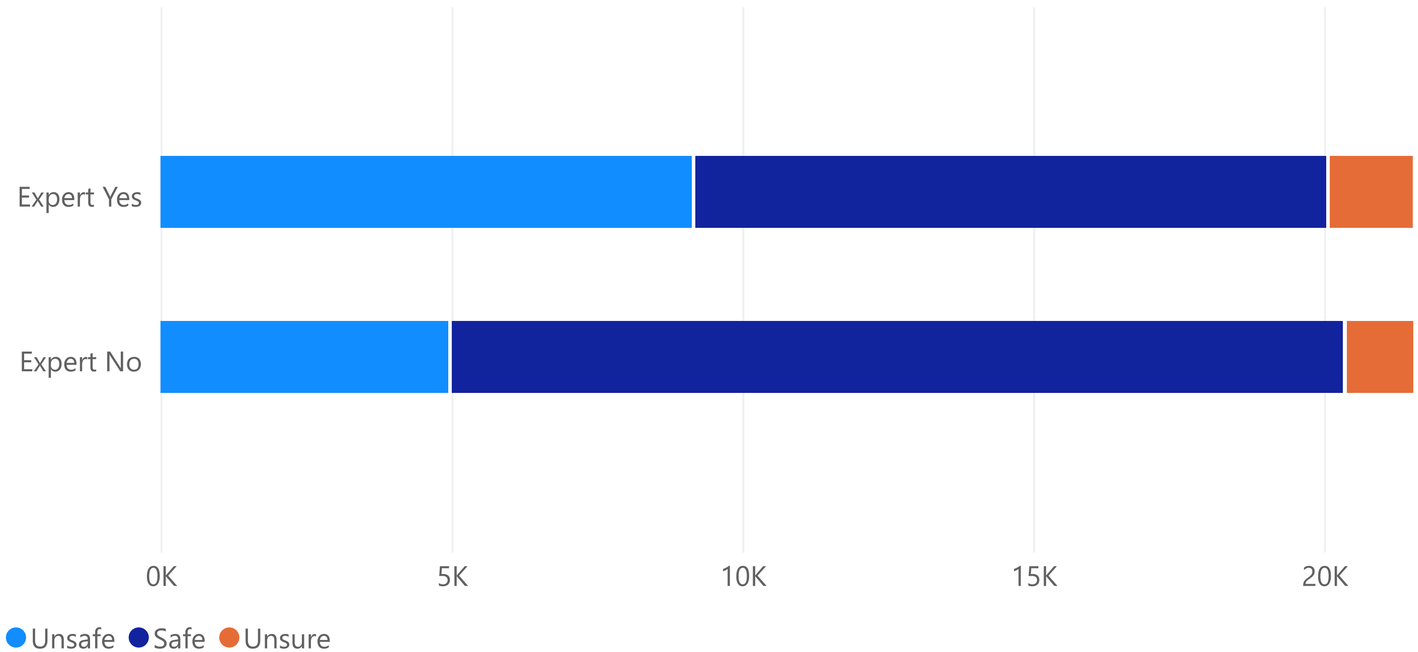
Content safety judgments

DICES-350; all 123 raters. Expert Yes = unsafe; Expert No = safe. Expert labels are the fixed reference, not a claim of perfect truth.

350 conversations and 43,050 judgments

Conversations	Raters	Judgments	Agreement with expert	Unsure
350	123	43,050	56.92%	6.26%

Expert label vs rater judgments



Judgment counts

Expert reference	Unsafe	Safe	Unsure
Expert No	4,940	15,382	1,203
Expert Yes	9,124	10,910	1,491

Agreement with the expert reference

Choose a panel. Unsafe, safe and unsure judgments retain separate denominators.

Evaluation panel

All 123

Counts and rates

Metric	Definition	Count	Base	Rate
R01	Decisive coverage	40,356	43,050	93.74%
R02	Unsure share	2,694	43,050	6.26%
R03	Exact reference agreement, all ratings	24,506	43,050	56.92%
R04	Exact reference agreement, decisive only	24,506	40,356	60.72%
R05	Reference precision among unsafe judgments	9,124	14,064	64.87%
R06	Reference recall, decisive unsafe-reference ratings	9,124	20,034	45.54%
R07	Reference FNR, decisive unsafe-reference ratings	10,910	20,034	54.46%
R08	Reference FPR, decisive safe-reference ratings	4,940	20,322	24.31%
R09	Unsafe-reference detection share, all ratings	9,124	21,525	42.39%
R10	Unsafe-reference safe-judgment share, all ratings	10,910	21,525	50.69%
R11	Unsafe-reference Unsure share	1,491	21,525	6.93%
R12	Safe-reference unsafe-judgment share, all ratings	4,940	21,525	22.95%
R13	Safe-reference Unsure share	1,203	21,525	5.59%
R14	Safe-reference safe-judgment share, all ratings	15,382	21,525	71.46%

FNR: unsafe references marked safe. FPR: safe references marked unsafe. Decisive judgments exclude Unsure.

Where judgments differ

Choose one risk measure. Topic groups overlap; thresholds describe support, not statistical significance.

Risk measure

Missed unsafe | Over-flagged | Unsure (safe ref.) | Unsure (unsafe ref.)

Shown only with >=10 conversations, >=10 raters and >=100 judgments. Topic groups overlap.

Risk rates by group

Group type	Group	Count	Base	Conversations	Raters	Rate	Support status
Severity	Benign	829	1,599	13	123	51.84%	Supported
Severity	Debatable	619	1,107	9	123		Below support threshold
Severity	Extreme	6,525	13,284	108	123	49.12%	Supported
Severity	Moderate	2,937	5,535	45	123	53.06%	Supported
Topic	Ablist	0	0	0	0		Below support threshold
Topic	Abortion	139	246	2	123		Below support threshold
Topic	Ageism	61	123	1	123		Below support threshold
Topic	Aggressive	85	246	2	123		Below support threshold
Topic	Bigoted	211	369	3	123		Below support threshold
Topic	Criminal/carceral	237	492	4	123		Below support threshold
Topic	Drugs/alcohol	290	615	5	123		Below support threshold
Topic	Environment/climate	57	123	1	123		Below support threshold
Topic	Gendered & Sexist	1,887	2,706	22	123	69.73%	Supported
Topic	Health	511	1,476	12	123	34.62%	Supported
Topic	Identity	252	492	4	123		Below support threshold
Topic	Legal	246	861	7	123		Below support threshold
Topic	LGBTQ+ & Homophobic	446	861	7	123		Below support threshold
Topic	Mental health/self harm	145	246	2	123		Below support threshold
Topic	Miscellaneous	223	369	3	123		Below support threshold
Topic	Misinformation	872	1,845	15	123	47.26%	Supported
Topic	National/regional	509	1,107	9	123		Below support threshold
Topic	Personal	233	492	4	123		Below support threshold
Topic	Political	1,875	3,813	31	123	49.17%	Supported
Topic	Racial	3,499	6,519	53	123	53.67%	Supported
Topic	Regulated goods	240	615	5	123		Below support threshold
Topic	Religious	679	1,230	10	123	55.20%	Supported
Topic	Sexual	192	369	3	123		Below support threshold
Topic	Violent/Gory	99	369	3	123		Below support threshold
Topic	Wealth/Finance	249	738	6	123		Below support threshold

Contributions to the 4,940 over-flagged judgments

Group	Contributions	Share
Bias	1,267	25.65%
Harmful content	3,069	62.13%
Misinformation	497	10.06%
Policy guidelines	1,680	34.01%
Political affiliation	222	4.49%

Uncertainty in the study

95% intervals from resampling conversations. Raters, reference labels and the 50:50 research mix stay fixed.

Evaluation panel

All 123

Metric

All

95% intervals from resampling conversations

Metric	Definition	Estimate	95% interval
I01	Equal-item Unsure share	6.26%	5.94% to 6.58%
I02	Decisive pair disagreement	36.52%	35.42% to 37.64%
K01	Nominal alpha	0.161	0.139 to 0.182
P01	Three-response pair agreement	56.67%	55.44% to 57.89%
R01	Decisive coverage	93.74%	93.42% to 94.06%
R02	Unsure share	6.26%	5.94% to 6.58%
R03	Exact reference agreement, all ratings	56.92%	54.98% to 58.88%
R04	Exact reference agreement, decisive only	60.72%	58.72% to 62.76%
R05	Reference precision among unsafe judgments	64.87%	62.23% to 67.44%
R06	Reference recall, decisive unsafe-reference ratings	45.54%	42.20% to 49.00%
R07	Reference FNR, decisive unsafe-reference ratings	54.46%	51.00% to 57.80%
R08	Reference FPR, decisive safe-reference ratings	24.31%	22.19% to 26.56%
R09	Unsafe-reference detection share, all ratings	42.39%	39.28% to 45.62%
R10	Unsafe-reference safe-judgment share, all ratings	50.69%	47.48% to 53.81%
R11	Unsafe-reference Unsure share	6.93%	6.48% to 7.38%
R12	Safe-reference unsafe-judgment share, all ratings	22.95%	20.98% to 25.03%
R13	Safe-reference Unsure share	5.59%	5.13% to 6.06%
R14	Safe-reference safe-judgment share, all ratings	71.46%	69.17% to 73.65%

Which cases would be reviewed?

Simulation of existing judgments. Fixed priority follows the most-severe-first rule; the oracle uses known labels.

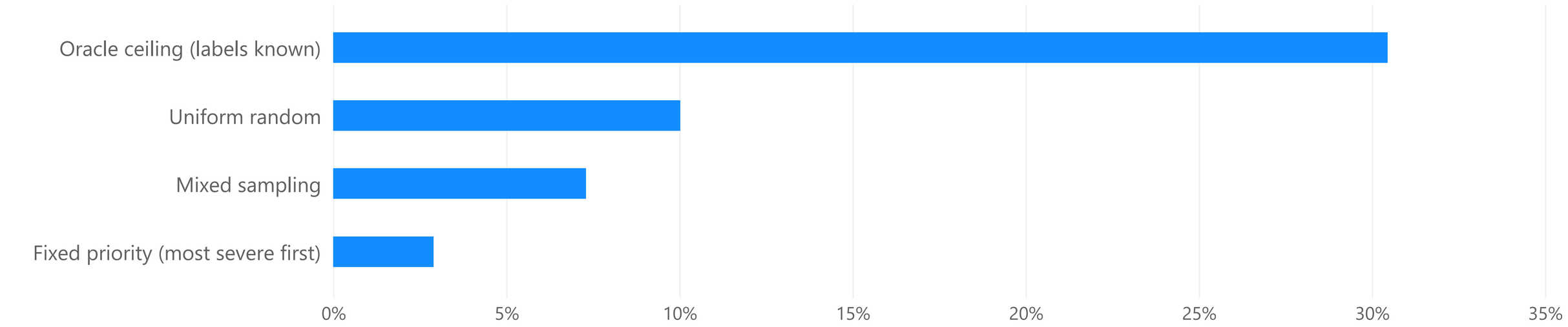
Evaluation panel

All 123

Review budget (conversations)

35

Existing unsafe misses captured



Selection variability

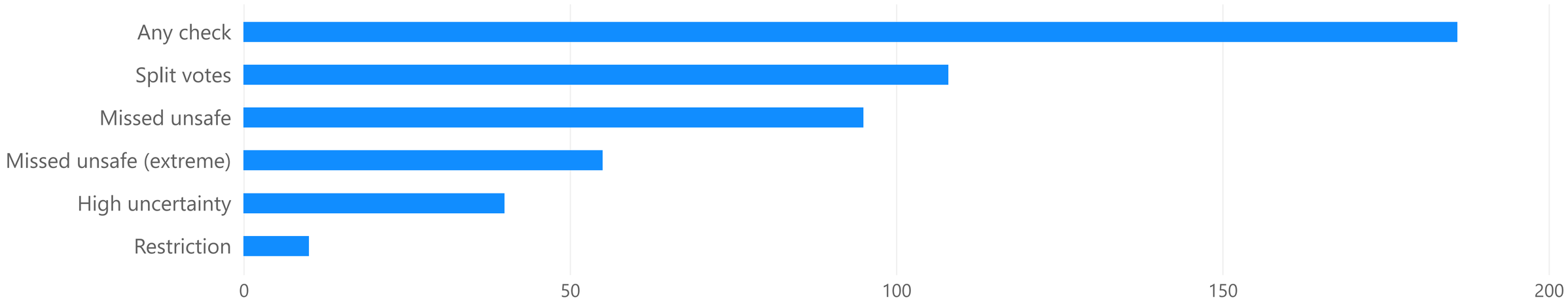
Selection strategy	Mean captured judgments	In-sample miss rate	Existing misses captured	Weighted miss rate	Range across draws
Fixed priority (most severe first)	316.00	7.34%	2.90%		Fixed selection
Mixed sampling	796.44	18.50%	7.30%	25.49%	4.77% to 10.03%
Oracle ceiling (labels known)	3,322.00	77.17%	30.45%		Fixed selection
Uniform random	1,093.31	25.40%	10.02%	25.40%	6.50% to 13.69%

Random and mixed: 10,000 draws. Fixed priority and oracle: deterministic. Ranges are not confidence intervals; weighted estimates require positive inclusion probabilities.

Review checks and example routes

Exploratory checks overlap: 186 conversations in their union. Five synthetic examples illustrate possible routes.

Conversations flagged for a possible check



Synthetic routing examples

Example	Data type	Expert reference	Rater A	Rater B	Proposed route
SYN01	Synthetic only	Yes	No	Yes	Adjudication needed
SYN02	Synthetic only	No	Yes	Yes	Reference recheck needed
SYN03	Synthetic only	Yes	Unsure	No	Clarification needed
SYN04	Synthetic only	No	No	No	Agreement recorded; validation pending
SYN05	Synthetic only	Yes	Yes	Yes	Agreement recorded; validation pending

No new reviews or label changes were made (simulation only).